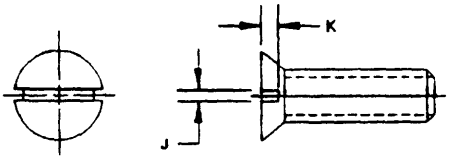


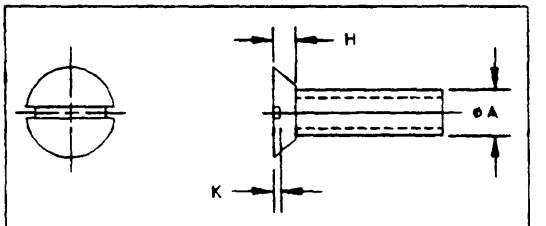
CROSS-RECESSED HEAD SCREW (DESIGN I)

THREAD SIZE	øA	øD			F	H REF		H1		J		K		R
		MAX SHARP	MIN SHARP	ABSOLUTE MIN WITH MAX F										
	MAX				MAX	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX
NO 2-56 UNC-2A	.086	172	156	150	.014	.051	.040	036	.028	.031	.023	.023	015	.035
NO 3-48 UNC-2A	099	199	181	175	015	059	048	041	034	035	.027	027	017	040
NO 4-40 UNC-2A	112	225	207	200	016	067	055	.047	.039	.039	.031	030	020	050
NO 5-40 UNC-2A	125	252	232	225	018	075	062	.053	.043	043	035	034	022	050
NO 6-32 UNC-2A	138	279	257	249	019	083	069	058	048	048	039	038	024	060
NO 8-32 UNC-2A	164	.332	308	300	022	100	084	070	059	054	045	045	029	060
NO 10-24 UNC-2A	190	385	359	348	025	116	.098	081	069	060	050	053	034	085
1/4-20 UNC-2A	.250	507	477	462	031	153	131	107	092	075	064	070	046	100
5/16-18 UNC-2A	313	635	600	581	037	191	165	134	116	084	072	088	058	110
3/8-16 UNC-2A	.375	762	722	700	043	.230	.200	161	140	094	081	106	070	125



SLOTTED HEAD SCREW (d)

SLOTTED HEAD SCREWS ARE INACTIVE FOR NEW DESIGN AFTER 17 FEB 59 - SEE NOTE 7



UNDER CUT DIMENSIONS-SLOTTED HEAD SCREW

SIZE	øA	H		K	
	MAX	MAX	MIN	MAX	MIN
NO 2-56UNC -2A	086	036	028	022	015
NO 3-48UNC -2A	.099	.041	.034	.025	017
NO 4-40UNC -2A	112	047	039	.028	.020
NO 5-40UNC -2A	125	053	043	.031	022
NO 6-32UNC -2A	.138	.058	.048	.035	.024
NO 8-32UNC -2A	164	070	059	040	029
NO 10-24UNC -2A	190	081	069	.047	.034
1/4 -20UNC -2A	250	107	092	061	046
5/16-18UNC -2A	313	134	116	076	058
3/8-16UNC -2A	375	161	140	091	070

(a) SEE REQUIREMENT 10
(b) SEE REQUIREMENT 6
(c) SEE REQUIREMENT 5
(d) SEE REQUIREMENT 7

REVIEWER SYI
ARMY-MI
AIR FORCE-

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AMSC N/A

APPROVED 15 MAY 43 REVISED (1) 15 JUN 50 (2) 21 FEB 52 (3) 17 FEB 59 (4) 30 JUN 94 (5) 31 MAY 95 (6) 22 APR 96

P A DLA-IS Other Cust ARMY-AV NAVY-AS AIR FORCE 99 PROCUREMENT SPECIFICATION FF S 92	AIR FORCE - NAVY AERONAUTICAL STANDARD SCREW, MACHINE, FLAT HEAD, 82 DEGREE, COARSE THREAD SUPERSEDES FORMER AIR FORCE AND NAVY STANDARD ISSUE OF AN505 AND USAF DRAWINGS 4284937 4285702	AN505 SHEET 1 OF 3

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FED SUP CLASS
5305

DASH NUMBERS FOR CARBON STEEL SCREWS (f)

L LENGTH		NO 2-56	NO 3-48	NO 4-40	NO 5-40	NO 6-32	NO 8-32	NO 10-24	1/4-20	5/16-18	3/8-16
3/16		2-3	3-3	4-3(d)	5-3(d)	6-3(d)					
1/4		2-4	3-4	4-4	5-4	6-4	8-4(d)				
5/16		2-5	3-5	4-5	5-5	6-5	8-5	10-5(d)	416-5(d)		
3/8		2-6	3-6	4-6	5-6	6-6	8-6	10-6	416-6(d)		
7/16		2-7	3-7	4-7	5-7	6-7	8-7	10-7	416-7(d)		
1/2	+0	2-8	3-8	4-8	5-8	6-8	8-8	10-8	416-8(d)	516-8(d)	616-8(d)
9/16	-1/32	2-9	3-9	4-9	5-9	6-9	8-9	10-9	416-9	516-9	616-9(d)
5/8		2-10	3-10	4-10	5-10	6-10	8-10	10-10	416-10	516-10	616-10
11/16		2-11	3-11	4-11	5-11	6-11	8-11	10-11	416-11	516-11	616-11
3/4		2-12	3-12	4-12	5-12	6-12	8-12	10-12	416-12	516-12	616-12
13/16			3-13	4-13	5-13	6-13	8-13	10-13	416-13	516-13	616-13
7/8			3-14	4-14	5-14	6-14	8-14	10-14	416-14	516-14	616-14
15/16			3-15	4-15	5-15	6-15	8-15	10-15	416-15	516-15	616-15
1			3-16	4-16	5-16	6-16	8-16	10-16	416-16	516-16	616-16
1-1/8				4-18	5-18	6-18	8-18	10-18	416-18	516-18	616-18
1-1/4				4-20	5-20	6-20	8-20	10-20	416-20	516-20	616-20
1-3/8				4-22	5-22	6-22	8-22	10-22	416-22	516-22	616-22
1-1/2	+0 -1/16			4-24	5-24	6-24	8-24	10-24	416-24	516-24	616-24
1-5/8						6-26	8-26	10-26	416-26	516-26	616-26
1-3/4						6-28	8-28	10-28	416-28	516-28	616-28
1-7/8						6-30	8-30	10-30	416-30	516-30	616-30
2						6-32	8-32	10-32	416-32	516-32	616-32
2-1/8						6-34	8-34	10-34	416-34	516-34	616-34
2-1/4						6-36	8-36	10-36	416-36	516-36	616-36
2-3/8						6-38	8-38	10-38	416-38	516-38	616-38
2-1/2	+0 -3/32					6-40	8-40	10-40	416-40	516-40	616-40
2-5/8						6-42	8-42	10-42	416-42	516-42	616-42
2-3/4						6-44	8-44	10-44	416-44	516-44	616-44
2-7/8						6-46	8-46	10-46	416-46	516-46	616-46
3						6-48	8-48	10-48	416-48	516-48	616-48

(d) SEE REQUIREMENT 7
(f) SEE REQUIREMENT 11

RATED STRENGTH IN POUNDS
(FOR REFERENCE PURPOSES ONLY)

SIZE	MINIMUM ULTIMATE TENSILE BREAKING STRENGTH AT ROOT DIA	MINIMUM SINGLE SHEAR STRENGTH AT ROOT DIA
NO 2-56	170	102
NO 3-48	225	135
NO 4-40	275	165
NO 5-40	370	222
NO 6-32	410	246
NO 8-32	660	396
NO 10-24	800	480
1/4-20	1480	890
5/16-18	2500	1500
3/8-16	3730	2240

ULTIMATE TENSILES BASED ON 55,000 PSI SHEAR STRENGTHS 60% OF ULTIMATE TENSILES

REQUIREMENTS

1. MATERIAL:
CARBON STEEL GRADE 1010 (UNS G10100) IN ACCORDANCE WITH ASTM A108, MINIMUM 55,000 UTS OR OTHER CARBON STEELS MEETING THE SAME PHYSICAL REQUIREMENTS
CORROSION RESISTANT STEEL IN ACCORDANCE WITH QQ-S-763, CLASS 430 (UNS S43000) OR UNS S30430 (CLASS XM-7), ASTM A582, TYPE 303 (UNS S30300) OR TYPE 303Se (UNS S30323) OR MIL-S-7720, COMPOSITION 302 (UNS S30200) FOR COLD UPSETTING PROCESS.
BRASS IN ACCORDANCE WITH ASTM B16, ASTM B36, ASTM B121, ASTM B134, ASTM B159 OR ASTM B206 WITH A MINIMUM ULTIMATE TENSILE STRENGTH OF 55,000 PSI ALUMINUM ALLOY 2024 (UNS A92024), IN ACCORDANCE WITH QQ-A-225/3 OR QQ-A-225/6

2 FINISH:
CARBON STEEL CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3
BRASS BLACK CHEMICAL FINISH IN ACCORDANCE WITH MIL-F-495 OR CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3, DEPENDING ON PART NUMBER CODING (SEE REQUIREMENT 11)
CORROSION RESISTANT STEEL PASSIVATE IN ACCORDANCE WITH QQ-P-35
ALUMINUM ALLOY ANODIZE IN ACCORDANCE WITH MIL-A-8625, TYPE I OR II, CLASS 1

3 HEAT TREATMENT
ALUMINUM ALLOY 62,000 PSI MINIMUM ULTIMATE TENSILE STRENGTH, IN ACCORDANCE WITH MIL-H-6088

4 THREADS:
THREADS SHALL BE CLASS 2A IN ACCORDANCE WITH FED-STD-H28/2 ACCEPTABILITY OF SCREW THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 21.

5 THREAD LENGTH:
FOR SCREWS UP TO AND INCLUDING 2 INCHES IN LENGTH, THE COMPLETE THREADS SHALL EXTEND TO WITHIN TWO (2) THREADS OF THE BEARING SURFACE OF THE HEAD, OR CLOSER IF PRACTICABLE SCREWS OF LONGER LENGTH SHALL HAVE A MINIMUM COMPLETE THREAD OF 1-3/4 INCHES.

6 UNTHREADED PORTION
THE DIAMETER OF THE UNTHREADED PORTION OF THE SCREW SHALL NOT BE LESS THAN THE MINIMUM PITCH DIAMETER NOR MORE THAN THE MAXIMUM MAJOR DIAMETER OF THE THREAD

7 UNDERCUT HEADS
DASH NUMBERS DESIGNATED BY (d) SHALL HAVE UNDERCUT HEADS

8 HEAD DRIVE STYLE
CROSS-RECESS RECESS IN ACCORDANCE WITH MS9006
SLOTTED HEAD REFER TO TABLE FOR DIMENSIONS (SEE NOTE 7)

REVIEWER SYMBOLS
ARMY-MI
AIR FORCE-82

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APPROVED 15 MAY 43 REVISED (4) 30 JUN 94 (5) 31 MAY 95 (6) 22 APR 96

P A DLA IS Other Cust ARMY-AV AIR FORCE 99	AIR FORCE NAVY AERONAUTICAL STANDARD SCREW, MACHINE, FLAT HEAD 82 DEGREE COARSE THREAD	AN505
PROCUREMENT SPECIFICATION FF-S-92	INTERIM FOR AIR FORCE AND NAVY STANDARD ISSUED AN505 AND USA DRAWING 42B493[illegible] 4 APR 02	SHEET 2 OF 3

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9 MANUFACTURER IDENTIFICATION.
SCREWS WITH NOMINAL SIZES #10 (1900) AND LARGER SHALL BE PERMANENTLY MARKED WITH A SYMBOL IDENTIFYING IT S MANUFACTURER MARKINGS SHALL BE PLACED ON THE TOP OF HEAD AND INDENTED METHOD AND SIZE OF MARKINGS SHALL CONFORM TO THE REQUIREMENTS FOR PERMANENT MARKINGS SPECIFIED IN SAE AS478

10 MATERIAL IDENTIFICATION.
CORROSION RESISTANT STEEL SCREWS WITH NOMINAL SIZES #5 (1250) AND LARGER SHALL BE PERMANENTLY MARKED WITH ONE DASH ("-") MARKINGS SHALL BE PLACED ON THE TOP OF HEAD AND INDENTED METHOD AND SIZE OF MARKINGS SHALL CONFORM TO THE REQUIREMENTS FOR PERMANENT MARKINGS SPECIFIED IN SAE AS478.

11 CODE
NO CODE BETWEEN FIRST AND SECOND DASH NUMBER = SLOTTED HEAD SCREW (SEE NOTE 7)
"R" BETWEEN FIRST AND SECOND DASH NUMBER = CROSS-RECESSED HEAD SCREW
NO CODE BEFORE FIRST DASH NUMBER = CARBON STEEL SCREW WITH CADMIUM PLATING
"B" BEFORE FIRST DASH NUMBER = BRASS SCREW WITH BLACK CHEMICAL FINISH
"PB" BEFORE FIRST DASH NUMBER = BRASS SCREW WITH CADMIUM PLATING
"UB" BEFORE FIRST DASH NUMBER = BRASS SCREW WITH NO PLATING OR FINISH
"C" BEFORE FIRST DASH NUMBER = CORROSION RESISTANT STEEL SCREW, PASSIVATED.
"DD" BEFORE FIRST DASH NUMBER = ALUMINUM ALLOY SCREW, ANODIZED

EXAMPLES

AN505-4-8 = SLOTTED HEAD SCREW, CARBON STEEL, CADMIUM PLATED, 4-40 UNC-2A THREAD, 1/2 INCH LONG

AN505-4R8 = CROSS-RECESSED HEAD SCREW, CARBON STEEL, CADMIUM PLATED, 4-40 UNC-2A THREAD, 1/2 INCH LONG

AN505B4-8 = SLOTTED HEAD SCREW, BRASS, BLACK CHEMICAL FINISH, 4-40 UNC-2A THREAD, 1/2 INCH LONG

AN505B4R8 = CROSS RECESSED HEAD SCREW, BRASS, BLACK CHEMICAL FINISH, 4-40 UNC-2A THREAD, 1/2 INCH LONG

AN505PB4-8 = SLOTTED HEAD SCREW, BRASS, CADMIUM PLATED, 4-40 UNC-2A THREAD, 1/2 INCH LONG

AN505PB4R8 = CROSS-RECESSED HEAD SCREW, BRASS, CADMIUM PLATED, 4-40 UNC-2A THREAD, 1/2 INCH LONG

AN505UB4-8 = SLOTTED HEAD SCREW, BRASS, UNFINISHED, 4-40 UNC-2A THREAD, 1/2 INCH LONG

AN505UB4R8 = CROSS-RECESSED HEAD SCREW, BRASS, UNFINISHED, 4-40 UNC-2A THREAD, 1/2 LONG

AN505C4-8 = SLOTTED HEAD SCREW, CORROSION RESISTANT STEEL, PASSIVATED, 4-40 UNC-2A THREAD, 1/2 INCH LONG

AN505C4R8 = CROSS RECESSED HEAD SCREW, CORROSION RESISTANT STEEL, PASSIVATED, 4-40 UNC-2A THREAD, 1/2 INCH LONG

AN505DD4-8 = SLOTTED HEAD SCREW, ALUMINUM ALLOY, ANODIZED, 4-40 UNC-2A THREAD, 1/2 INCH LONG

AN505DD6R8 = CROSS-RECESSED SCREW, ALUMINUM ALLOY, ANODIZED, 6-32 UNC-2A THREAD, 1/2 INCH LONG

NOTES.

1 ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED

2 INTERPRET DIMENSIONS AND TOLERANCES IN ACCORDANCE WITH ANSI/ASME Y14.5M

3 SCREWS SHALL BE FREE OF ALL LOOSE HANGING BURRS OR SLIVERS WHICH MAY BECOME DISLODGED DURING USAGE

4 DO NOT USE UNASSIGNED DASH NUMBERS OR CODES

5 IN CASE OF CONFLICT, THE REQUIREMENTS SPECIFIED HEREIN SHALL TAKE PRECEDENCE OVER THE REQUIREMENTS SPECIFIED IN SPECIFICATION FF-S-92 FOR DIMENSIONS AND TOLERANCES NOT SPECIFIED HEREIN OR FF-S-92, REFER TO ANSI B18 6 3

6 THESE SCREWS ARE NOT TO BE USED IN THE CONSTRUCTION OF AIRFRAMES OR TO ATTACH EQUIPMENT TO THE AIRFRAME

7 SLOTTED HEAD SCREWS ARE INACTIVE FOR NEW DESIGN AFTER 17 FEB 59 AN505 IS SUPERSEDED IN PART BY MS35190, MS35198, MS35202, AND MS51959

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AMSC N/A

PA DLA-IS Other Cust ARMY-AV NAVY-AS AIR FORCE-99 PROCUREMENT SPECIFICATION FF-S-92	AIR FORCE - NAVY AERONAUTICAL STANDARD SCREW MACHINE FLAT HEAD 82 DEGREE COARSE THREAD SUPERSEDES FORMER AIR FORCE AND NAVY STANDARD ISSUE OF AN505 AND USAF DRAWINGS [illegible]	AN505 SHEET 3 OF 3

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